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JEE Main Physics Practice Sheet — DPP-031

Topic: Dynamics of Motion on Rough Inclined Planes

Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A block of mass $m = 4$ kg is released from rest from the top of an inclined plane of length $L = 10$ m and inclination $\theta = 37°$. If the coefficient of kinetic friction is $\mu_k = 0.25$, the speed of the block upon reaching the bottom is ($\sin 37° = 0.6, \cos 37° = 0.8, g = 10$ m/s^2):

(A) 8.94 m/s

(B) 10.0 m/s

(C) 6.32 m/s

(D) 12.0 m/s

Q2. An inclined plane of inclination θ has its upper half completely smooth and lower half rough with coefficient of kinetic friction μ. A body released from rest at the top just comes to rest at the bottom of the plane. The value of μ is:

(A) $\tan\theta$

(B) $2\tan\theta$

(C) $\frac{1}{2}\tan\theta$

(D) $4\tan\theta$

Q3. A block of mass m is projected up a rough inclined plane of angle $\theta = 30°$ with initial velocity u. It comes to rest after travelling a distance s up the incline. If $\mu_k = \frac{1}{2\sqrt{3}}$, the distance s travelled along the plane is ($g = 10$ m/s^2):

(A) $\frac{u^2}{15}$

(B) $\frac{u^2}{10}$

(C) $\frac{u^2}{20}$

(D) $\frac{2u^2}{15}$

Q4. A block of mass $m = 2$ kg is projected up a rough inclined plane of inclination $\theta = 37°$ with speed $v_0 = 10$ m/s. If $\mu_k = 0.5$, the speed with which the block returns back to the point of projection is:

(A) 4.47 m/s

(B) 5.0 m/s

(C) 7.07 m/s

(D) 2.24 m/s

Q5. The minimum force F applied parallel to a rough inclined plane of angle θ required to just pull a block of mass m up the plane is F_1, and the minimum force required to prevent it from sliding down is F_2. If $\theta > \tan^{-1}\mu_s$, then $(F_1 - F_2)$ is equal to:

(A) $2\mu_s mg\cos\theta$

(B) $2mg\sin\theta$

(C) $\mu_s mg\cos\theta$

(D) $mg(\sin\theta + \mu_s\cos\theta)$

Q6. A block of mass m slides down an inclined plane of inclination θ with variable friction coefficient given by $\mu(x) = kx$, where x is the distance travelled from the top rest position. The maximum distance $x_{\max}$ travelled by the block before it stops is:

(A) $\frac{2\tan\theta}{k}$

(B) $\frac{\tan\theta}{k}$

(C) $\frac{\tan\theta}{2k}$

(D) $\frac{4\tan\theta}{k}$

Q7. In Question 6, the position x where the block achieves its maximum velocity is:

(A) $\frac{\tan\theta}{k}$

(B) $\frac{2\tan\theta}{k}$

(C) $\frac{\tan\theta}{2k}$

(D) $\frac{\tan\theta}{\sqrt{k}}$

Q8. A block of mass $m = 5$ kg rests on a rough inclined plane of inclination $\theta = 30°$ ($\mu_s = 0.7$). The magnitude of the frictional force acting on the block is ($g = 10$ m/s^2):

(A) 25.0 N

(B) 30.3 N

(C) 35.0 N

(D) 50.0 N

Q9. A block of mass m is pushed up a rough inclined plane of inclination θ by a horizontal force F. The normal reaction N exerted by the incline on the block is:

(A) $mg\cos\theta + F\sin\theta$

(B) $mg\cos\theta - F\sin\theta$

(C) $mg\cos\theta$

(D) $(mg + F)\cos\theta$

Q10. In Question 9, the minimum horizontal force F required to just move the block of mass m up the rough inclined plane (μ_s) is:

(A) $mg\left[\frac{\sin\theta + \mu_s\cos\theta}{\cos\theta - \mu_s\sin\theta}\right]$

(B) $mg\left[\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta}\right]$

(C) $mg(\tan\theta + \mu_s)$

(D) $\dfrac{\mu_s mg}{\cos\theta - \mu_s \sin\theta}$

Q11. A block is placed on a rough inclined plane whose inclination θ can be varied. The time taken by the block to slide down a fixed length L of the plane is minimum when the angle of inclination θ satisfies:

(A) $\tan 2\theta = -\dfrac{1}{\mu_k}$

(B) $\tan\theta = \mu_k$

(C) $\tan 2\theta = \mu_k$

(D) $\theta = 45° + \dfrac{1}{2}\tan^{-1}\mu_k$

Q12. A body of mass m is released from height h from rest on a rough inclined plane of angle θ and base length b. The work done by friction as the body reaches the bottom of the incline is:

(A) $-\mu_k mgb$

(B) $-\mu_k mgh$

(C) $-\mu_k mg\sqrt{h^2+b^2}$

(D) $-\mu_k mg\left(\dfrac{h}{\sin\theta}\right)$

Q13. Two blocks of masses $m_1 = 2$ kg and $m_2 = 4$ kg connected by an inextensible light string slide down an inclined plane of angle $\theta = 37°$. If the coefficient of friction for m_1 (upper block) is $\mu_1 = 0.4$ and for m_2 (lower block) is $\mu_2 = 0.2$, the tension in the string during downward motion is ($g = 10$ m/s^2):

(A) 2.13 N

(B) 0 N

(C) 4.27 N

(D) 1.07 N

Q14. In Question 13, if the positions of the two blocks are reversed so that m_2 ($\mu_2 = 0.2$) is above m_1 ($\mu_1 = 0.4$), the tension in the connecting string becomes:

(A) 0 N

(B) 2.13 N

(C) 4.27 N

(D) 6.40 N

Q15. A block is released from rest at the top of a rough incline ($\mu_k = 0.25, \theta = 45°$). If the length of the incline is 16 m, the time taken by the block to slide down is ($g = 9.8$ m/s^2):

(A) 2.48 s

(B) 3.50 s

(C) 1.80 s

(D) 4.00 s

Q16. A block of mass m slides down a rough curved ramp of height h and arrives at the bottom with speed $v = \sqrt{gh}$. The work done by friction along the curve is:

(A) $-\dfrac{1}{2}mgh$

(B) $-mgh$

(C) $-\dfrac{3}{2}mgh$

(D) $-\dfrac{1}{4}mgh$

Q17. A triangular wedge of mass M and inclination θ rests on a smooth horizontal floor. A block of mass m is placed on its rough face (μ). The maximum horizontal acceleration a of the wedge to the right so that the block does not slide up the incline is:

(A) $g\left[\dfrac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}\right]$

(B) $g\left[\dfrac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta}\right]$

(C) $g\tan(\theta + \lambda)$

(D) Both (A) and (C)

Q18. A block is projected up an incline with speed u. After reaching the highest point, it slides down. If the ratio of kinetic energy at return to initial kinetic energy is η ($\eta < 1$), the coefficient of kinetic friction μ_k is:

(A) $\tan\theta\left(\dfrac{1-\eta}{1+\eta}\right)$

(B) $\tan\theta\left(\dfrac{1-\sqrt{\eta}}{1+\sqrt{\eta}}\right)$

(C) $\tan\theta\left(\dfrac{1+\eta}{1-\eta}\right)$

(D) $\tan\theta(1-\eta)$

Q19. A block of mass $m = 1$ kg is kept on an inclined plane of inclination $\theta = 45°$ ($\mu_s = 0.8$). A force F is applied horizontally to keep the block in equilibrium. The range of F for static equilibrium is ($g = 10$ m/s^2):

(A) $\dfrac{10}{9}$ N $\leq F \leq 90$ N

(B) 1.11 N $\leq F \leq 90$ N

(C) 2.0 N $\leq F \leq 80$ N

(D) Both (A) and (B) are equivalent

Q20. A wooden block of mass m resting on an incline θ is connected to a hanging mass M via a light string passing over a frictionless pulley at the top. The minimum mass M required to just prevent the block m from sliding down the incline ($\mu_s < \tan\theta$) is:

(A) $m(\sin\theta - \mu_s\cos\theta)$

(B) $m(\sin\theta + \mu_s\cos\theta)$

(C) $\mu_s m\cos\theta$

(D) $m\sin\theta$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-031)

Dynamics of Motion on Rough Inclined Planes

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	D	16	A
2	B	7	A	12	A	17	D
3	A	8	A	13	A	18	A
4	A	9	A	14	A	19	D
5	A	10	A	15	A	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (A)
Downhill acceleration along the rough incline:

$$a = g(\sin\theta - \mu_k \cos\theta)$$

$$a = 10(0.6 - 0.25 \times 0.8) = 10(0.6 - 0.20) = 4.0 \text{ m/s}^2$$

Using kinematic formula $v^2 = u^2 + 2aL$ with $u = 0$:

$$v = \sqrt{2 \times 4.0 \times 10} = \sqrt{80} \approx 8.94 \text{ m/s}$$

Sol 2. Option (B)
Let the total length of the incline be $2s$.
Work-Energy Theorem from top to bottom ($\Delta K = 0$):

$$W_{\text{gravity}} + W_{\text{friction}} = 0$$

$$mg(2s)\sin\theta - (\mu mg\cos\theta)s = 0$$

$$2mgs\sin\theta = \mu mgs\cos\theta \implies \mu = 2\tan\theta$$

Sol 3. Option (A)
During upward motion, both gravity and kinetic friction decelerate the block:

$$a_{\text{up}} = g(\sin 30^\circ + \mu_k \cos 30^\circ) = 10\left(\frac{1}{2} + \frac{1}{2\sqrt{3}} \cdot \frac{\sqrt{3}}{2}\right)$$

$$a_{\text{up}} = 10\left(\frac{1}{2} + \frac{1}{4}\right) = 10\left(\frac{3}{4}\right) = 7.5 \text{ m/s}^2$$

Using $v^2 = u^2 - 2a_{\text{up}}s$ with $v = 0$:

$$s = \frac{u^2}{2a_{\text{up}}} = \frac{u^2}{2(7.5)} = \frac{u^2}{15}$$

Sol 4. Option (A)
Upward deceleration:

$$a_{\text{up}} = g(\sin 37^\circ + \mu_k \cos 37^\circ) = 10(0.6 + 0.5 \times 0.8) = 10(1.0) = 10 \text{ m/s}^2$$

Distance travelled up: $s = \frac{v_0^2}{2a_{\text{up}}} = \frac{100}{2(10)} = 5$ m.
Downward acceleration:

$$a_{\text{down}} = g(\sin 37^\circ - \mu_k \cos 37^\circ) = 10(0.6 - 0.4) = 2.0 \text{ m/s}^2$$

Return speed at bottom:

$$v = \sqrt{2a_{\text{down}}s} = \sqrt{2 \times 2.0 \times 5} = \sqrt{20} \approx 4.47 \text{ m/s}$$

Sol 5. Option (A)
To pull up: $F_1 = mg\sin\theta + \mu_s mg\cos\theta$.
To prevent sliding down: $F_2 = mg\sin\theta - \mu_s mg\cos\theta$.
Difference:

$$F_1 - F_2 = 2\mu_s mg\cos\theta$$

Sol 6. Option (A)
Equation of motion along the incline:

$$v\frac{dv}{dx} = g\sin\theta - \mu(x)g\cos\theta = g(\sin\theta - kx\cos\theta)$$

Integrating from $x = 0$ ($v = 0$) to $x = x_{\max}$ ($v = 0$):

$$\int_0^{x_{\max}} g(\sin\theta - kx\cos\theta)\,dx = 0$$

$$x_{\max}\sin\theta - \frac{1}{2}kx_{\max}^2\cos\theta = 0 \implies x_{\max} = \frac{2\tan\theta}{k}$$

Sol 7. Option (A)
Maximum velocity occurs when acceleration is zero:

$$a = g(\sin\theta - kx\cos\theta) = 0 \implies kx\cos\theta = \sin\theta \implies x = \frac{\tan\theta}{k}$$

Sol 8. Option (A)
Angle of repose $\phi = \tan^{-1}(\mu_s) = \tan^{-1}(0.7) \approx 35^\circ$.
Since $\theta = 30^\circ < 35^\circ$, the block is in static equilibrium and does not slide.
Static friction balances the parallel gravity component:

$$f_s = mg\sin 30^\circ = 5 \times 10 \times 0.5 = 25.0 \text{ N}$$

Sol 9. Option (A)
Resolving forces perpendicular to the incline:

$$N = mg\cos\theta + F\sin\theta$$

Sol 10. Option (A)
For impending motion up the incline, resolving along the plane:

$$F\cos\theta = mg\sin\theta + f_s^{\max} = mg\sin\theta + \mu_s(mg\cos\theta + F\sin\theta)$$

$$F(\cos\theta - \mu_s\sin\theta) = mg(\sin\theta + \mu_s\cos\theta)$$

$$F = mg\left[\frac{\sin\theta + \mu_s\cos\theta}{\cos\theta - \mu_s\sin\theta}\right]$$

Sol 11. Option (D)
Acceleration $a = g(\sin\theta - \mu_k\cos\theta)$.
Time to slide distance L: $t = \sqrt{\frac{2L}{a}} \propto (\sin\theta - \mu_k\cos\theta)^{-1/2}$.
To minimize t, maximize $f(\theta) = \sin\theta - \mu_k\cos\theta$.
$\frac{df}{d\theta} = \cos\theta + \mu_k\sin\theta = 0 \implies \tan\theta = -\frac{1}{\mu_k}$.

For a fixed vertical height h ($L = h/\sin\theta$), minimizing t gives $\theta = 45^\circ + \frac{1}{2}\tan^{-1}\mu_k$.

Sol 12. Option (A)
Work done by friction:

$$W_f = -f_k \cdot L = -(\mu_k mg\cos\theta)\left(\frac{b}{\cos\theta}\right) = -\mu_k mgb$$

where $b = L\cos\theta$ is the horizontal base distance.

Sol 13. Option (A)
Free-body equations for both blocks moving with common acceleration a: 1. $m_1 g\sin\theta - \mu_1 m_1 g\cos\theta + T = m_1 a$ 2. $m_2 g\sin\theta - \mu_2 m_2 g\cos\theta - T = m_2 a$
Adding both:

$$(m_1+m_2)g\sin\theta - (\mu_1 m_1 + \mu_2 m_2)g\cos\theta = (m_1+m_2)a$$

$$6(10)(0.6) - (0.4\times2+0.2\times4)(10)(0.8) = 6a \implies 36 - 12.8 = 6a \implies a = 3.867\ \text{m/s}^2$$

From (1): $T = m_1 a - m_1 g\sin\theta + \mu_1 m_1 g\cos\theta = 2(3.867) - 12 + 6.4 = 7.733 - 5.6 = 2.13$ N.

Sol 14. Option (A)
When lower block has higher friction ($\mu_1 = 0.4$) and upper block has lower friction ($\mu_2 = 0.2$), the upper block tends to accelerate faster than the lower block. As a result, the connecting string becomes slack ($T = 0$ N) and both blocks slide in contact or independently.

Sol 15. Option (A)
Acceleration down the 45° plane:

$$a = g(\sin 45^\circ - \mu_k\cos 45^\circ) = 9.8\left(\frac{1}{\sqrt{2}} - \frac{0.25}{\sqrt{2}}\right) = \frac{9.8\times0.75}{\sqrt{2}} = 5.197\ \text{m/s}^2$$

Using $L = \frac{1}{2}at^2$:

$$t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2\times16}{5.197}} = \sqrt{6.157} \approx 2.48\ \text{s}$$

Sol 16. Option (A)
By Work-Energy Theorem:

$$W_{\text{gravity}} + W_{\text{friction}} = \Delta K$$

$$mgh + W_f = \frac{1}{2}mv^2 - 0 = \frac{1}{2}m(gh) = \frac{1}{2}mgh$$

$$W_f = \frac{1}{2}mgh - mgh = -\frac{1}{2}mgh$$

Sol 17. Option (D)
In the non-inertial frame of the accelerating wedge (pseudo force ma to left): Normal force: $N = mg\cos\theta + ma\sin\theta$.
To prevent slipping up: $ma\cos\theta \le mg\sin\theta + \mu N = mg\sin\theta + \mu(mg\cos\theta + ma\sin\theta)$.

$$ma(\cos\theta - \mu\sin\theta) \le mg(\sin\theta + \mu\cos\theta) \implies a \le g\left[\frac{\sin\theta+\mu\cos\theta}{\cos\theta-\mu\sin\theta}\right]$$

Both forms are identical.

Sol 18. Option (A)
Let incline length be s.
Initial KE $K_i = \frac{1}{2}mu^2 = mgs(\sin\theta + \mu_k\cos\theta)$.
Return KE $K_f = mgs(\sin\theta - \mu_k\cos\theta)$.

$$\eta = \frac{K_f}{K_i} = \frac{\sin\theta - \mu_k\cos\theta}{\sin\theta + \mu_k\cos\theta} = \frac{\tan\theta - \mu_k}{\tan\theta + \mu_k}$$

$$\eta(\tan\theta + \mu_k) = \tan\theta - \mu_k \implies \mu_k(1+\eta) = \tan\theta(1-\eta) \implies \mu_k = \mathrm{ta}$$

Sol 19. Option (D)
Minimum F (preventing sliding down):

$$F_{\min} = mg\left[\frac{\sin 45^\circ - \mu_s\cos 45^\circ}{\cos 45^\circ + \mu_s\sin 45^\circ}\right] = 10\left[\frac{1-0.8}{1+0.8}\right] = 10\left(\frac{0.2}{1.8}\right) = \frac{10}{9}$$

Maximum F (preventing sliding up):

$$F_{\max} = mg\left[\frac{\sin 45^\circ + \mu_s\cos 45^\circ}{\cos 45^\circ - \mu_s\sin 45^\circ}\right] = 10\left[\frac{1+0.8}{1-0.8}\right] = 10\left(\frac{1.8}{0.2}\right) = 90$$

Hence $\frac{10}{9}$ N $\le F \le 90$ N.

Sol 20. Option (A)
For the block to stay at rest without slipping down, the string tension $T = Mg$ must satisfy:

$$T + f_s^{\max} \ge mg\sin\theta \implies Mg + \mu_s mg\cos\theta \ge mg\sin\theta$$

$$M_{\min} = m(\sin\theta - \mu_s\cos\theta)$$